

Chenshuo (Aiden) Wang

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EDUCATION

University of Minnesota – Twin Cities

Expected: May 2028

M.S. Mechanical Engineering

University of Minnesota – Twin Cities

Graduated

B.S. Mechanical Engineering

GPA: 3.6/4.0

EXPERIENCE

AltoAuto | *PyMAPDL, API, CAE Automation, Surrogate Modeling*

Founder / CAE Automation Engineer

Nov 2024 – Present

- Automated end-to-end CAD to FEA pipeline, deployed to FSAE and Solar Car team
- Built CAD automation extracting geometry features and machining parameters via SolidWorks API
- Developed automated meshing workflows with geometry cleanup, defeaturing, and element quality enforcement

Daikin Industries | *Thermal Modeling, Transient Analysis, Hardware Validation*

Modeling Engineer

Sep 2025 – Jan 2026

- Modeled two-phase liquid cooling loop transient dynamics under variable heat load
- Engineered liquid cooling circuits, sizing pump flow rates across worst-case thermal operating margins
- Developed DOE calibration harness sweeping UA, NTU, and pressure drop across valve operating envelope
- Calibrated R-32 two-phase model against hardware bench data, matching across the full operating range

Cold Plate Design | *Cold-Plate Optimization, CFD, Chip Cooling*

Simulation Engineer (IEEE ITherm Student Competition)

Jan 2025 – Present

- Optimized cold plate thermal resistance and pressure drop for a non-uniform chip power map.
- Performing conjugate heat transfer CFD in Ansys Fluent to predict junction temperature, pressure drop, and coolant flow
- Developing analytical thermal models (thermal resistance network, heat transfer correlations) to validate CFD results

Thermal-Fluids Research Lab | *Two-Phase Cooling, DAQ, Chip Assembly*

Research Assistant

Jan 2026 – Present

- Investigating phase-change cooling for high-power semiconductors to maximize chip-scale heat flux removal.
- Instrumented chip-level test articles with soldered thermocouple arrays and built a DAQ pipeline for Critical Heat Flux analysis.
- Correlated high-speed visualization of surface bubble dynamics with boiling curve heat transfer performance.

SKILLS

Core Mechanical: Thermodynamics, Fluid Mechanics, Heat Transfer, Solid Mechanics, GD&T (ASME Y14.5), Dynamics & Kinematics, Liquid Cooling Systems, Two-Phase Flow, Thermal Management

Computational Engineering: Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Turbulence Modeling, Meshing, Model-Based Design (MBD)

Mechatronics & Control: PID Control, Hardware-in-the-Loop (HIL), System ID, Real-Time DAQ, Electromechanical Actuation, MPC, LQR, EKF, Kalman Filter

Modeling & Simulation: ANSYS (Mechanical, Fluent, Icepak), Simulink, Simscape, SolidWorks, Creo, OpenFOAM, Code_Aster, Gmsh, RoboDK, Onshape

Programming & Scripting: Python, C/C++, MATLAB, PLC, Git, PyMAPDL, PyFluent, PyVista, SolidWorks API, PyTorch, FORTRAN